

$$x = ay^2(x) + by(x) + c. \quad 1 = 2ayy' + by'. \quad b = \frac{1 - 2ayy'}{y'}.$$

$$0 = \frac{\left(-2a(y')^2 - 2ayy''\right)y' - (1 - 2ayy')y''}{(y')^2}; \quad -2a(y')^3 - 2ayy'y'' - y'' + 2ayy'y'' = 0.$$

$$2a(y')^3 + y'' = 0. \quad a = -\frac{y''}{2(y')^3}. \quad 0 = -\frac{2y'''(y')^3 - 6(y'y'')^2}{2(y')^3}. \quad y'y''' = 3(y'')^2.$$